

A Two-Mode Jaynes–Cummings Model of Endogenous and Exogenous Phase Coherence: Falsifiable Predictions for Neural Dynamics

Miguel Angel Percudani^{*1} and Jorge Iván Díaz¹

¹Independent Research Collaboration

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Abstract

We present a two-mode quantum model of coupled neural dynamics in which an analytic sequential mode (a qubit) interacts with a non-analytic substrate mode (a truncated harmonic oscillator) through a time-dependent Jaynes–Cummings interaction. Dissipation is treated in the Lindblad formalism, and the coupling strength is modulated by an empirically accessible signal identified with the Default Mode Network activity. We show that the naive threshold $g_{\text{crit}} = \delta/2$, previously proposed as a lucidity criterion, is not a dynamical transition: it is a geometric label of the mixing angle. The true control structure is a two-dimensional phase diagram in the plane $(\alpha = g/\delta, \gamma_\phi)$. Numerical simulations show that the boundary between analytic-dominated and substrate-dominated steady states is a curve $g_{\text{cruce}}(\gamma_\phi)$, and that two coupling regimes with equal ratio $\mathcal{R} = g/(\gamma_\phi + \gamma_a)$ can produce qualitatively distinct asymptotic states. We propose a concrete falsifiable prediction for EEG/fMRI: the difference between endogenous (structured) and exogenous (forced) substrate access is not in the coupling strength but in the steady-state substrate excitation.

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^{*}Corresponding author: miguel_percudani@yahoo.com.ar. ORCID: [0009-0007-1748-3212](https://orcid.org/0009-0007-1748-3212).

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1 Introduction

The idea that cortical dynamics can be described in terms of coupled oscillators with phase-dependent interaction has a long history in computational neuroscience [1, 2]. More recently, quantum-optical formalisms have been proposed as effective descriptions of coherent neural population activity [3, 4]. We do not take a position on whether the brain is fundamentally quantum. We treat the Jaynes–Cummings formalism as a *phenomenological* model of two coexisting dynamical modes and explore its consequences.

The paper is organized as follows. Section 2 defines the model. Section 3 presents the numerical results. Section 4 states falsifiable predictions. Section 5 discusses the implications. Section 6 concludes.

2 Model

2.1 Hilbert space

We consider a bipartite quantum system

$$\mathcal{H} = \mathcal{H}_0 \otimes \mathcal{H}_1, \quad \dim \mathcal{H} = 2N_0, \quad (1)$$

with

- $\mathcal{H}_1 = \mathbb{C}^2$: the *analytic mode* (qubit). Basis $\{|0\rangle_1, |1\rangle_1\}$, where $|0\rangle_1$ is the *inactive* state and $|1\rangle_1$ is the *active* state.
- $\mathcal{H}_0 = \ell^2(\mathbb{N})$, truncated to N_0 levels: the *substrate mode* (oscillator). Fock basis $\{|n\rangle_0\}_{n=0}^{N_0-1}$.

2.2 Free Hamiltonians

The free Hamiltonians are

$$H_0 = \hbar\omega_0 a^\dagger a, \quad H_1 = \hbar\omega_1 \sigma_+ \sigma_-, \quad (2)$$

where $a, a^\dagger$ are oscillator operators, $\sigma_\pm$ are Pauli raising/lowering operators, and

$$\delta \equiv \omega_1 - \omega_0 \quad (3)$$

is the detuning.

2.3 Interaction

We use the standard Jaynes–Cummings coupling in the Rotating Wave Approximation:

$$H_{\text{int}} = \hbar g(t) \left(a^\dagger \sigma_- + a \sigma_+ \right), \quad (4)$$

with

$$\sigma_- = |0\rangle_1 \langle 1|, \quad \sigma_+ = |1\rangle_1 \langle 0|. \quad (5)$$

This coupling conserves the total excitation number $\mathcal{N} = a^\dagger a + \sigma_+ \sigma_-$.

2.4 DMN modulation

The coupling strength is modulated by the Default Mode Network (DMN) activity $A_{\text{DMN}}(t) \in [0, 1]$:

$$g(t) = g_0 \eta(A_{\text{DMN}}(t)), \quad \eta(A) = \frac{1}{1 + e^{\lambda(A - A_{\text{th}})}} \quad (6)$$

where g_0 is the maximal coupling, λ is the sigmoid stiffness, and A_{th} is the threshold.

The DMN activity evolves as an Ornstein–Uhlenbeck process:

$$\tau_{\text{DMN}} \frac{dA_{\text{DMN}}}{dt} = -(A_{\text{DMN}} - A_0) + \sigma \xi(t), \quad (7)$$

with $\langle \xi(t) \xi(t') \rangle = 2\delta(t - t')$.

2.5 Dissipation

The Lindblad dissipator is

$$\begin{aligned} \mathcal{L}_{\text{dec}}[\rho] = & \gamma_\phi (\sigma_z \rho \sigma_z - \rho) \\ & + \gamma_a (n_{\text{th}} + 1) \left(a \rho a^\dagger - \frac{1}{2} \{a^\dagger a, \rho\} \right) \\ & + \gamma_a n_{\text{th}} \left(a^\dagger \rho a - \frac{1}{2} \{a a^\dagger, \rho\} \right), \end{aligned} \quad (8)$$

with $\sigma_z = |1\rangle_1 \langle 1| - |0\rangle_1 \langle 0|$.

The master equation is

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [H, \rho] + \mathcal{L}_{\text{dec}}[\rho]. \quad (9)$$

2.6 Rabi frequency and control parameters

At fixed excitation manifold n , the Rabi frequency is

$$\Omega_R = \sqrt{g^2 + (\delta/2)^2}. \quad (10)$$

We define two dimensionless control parameters:

$$\alpha \equiv \frac{g}{\delta}, \quad \mathcal{R} \equiv \frac{g}{\gamma_\phi + \gamma_a}. \quad (11)$$

The mixing angle is

$$\theta_n = \frac{1}{2} \arctan \left(\frac{2g\sqrt{n+1}}{\delta} \right). \quad (12)$$

The naive label $g_{\text{crit}} = \delta/2$ corresponds to $\theta_0 = \pi/8$ and is *not* a dynamical threshold.

3 Numerical results

3.1 Setup

We integrate the Lindblad master equation with the BDF implicit solver of `scipy.integrate.solve_ivp`, with tolerances `rtol` = 10^{-7} and `atol` = 10^{-9} . Reference parameters: $\omega_0 = 6$ Hz, $\omega_1 = 40$ Hz, $\delta = 34$ Hz, $\gamma_a = 0.1$ Hz, $n_{\text{th}} = 0.05$.

The initial state is $|0\rangle_0 \otimes |1\rangle_1$ (substrate vacuum, analytic mode active). Note that $|0\rangle_0 \otimes |0\rangle_1$ is a *dark state* of the standard Jaynes–Cummings coupling and must be avoided as initial condition.

3.2 Rabi oscillation

For $g = 17$ Hz, $\gamma_\phi = 0.5$ Hz, the Rabi frequency is $\Omega_R = 24.04$ Hz. Figure 1 shows 10 periods of the normalized evolution. The envelope decays at rate $\sim \gamma_a/2 + \gamma_\phi$, and the mean drifts toward $\langle \sigma_z \rangle < 0$ as the initial spin excitation is transferred to the cavity and dissipated.

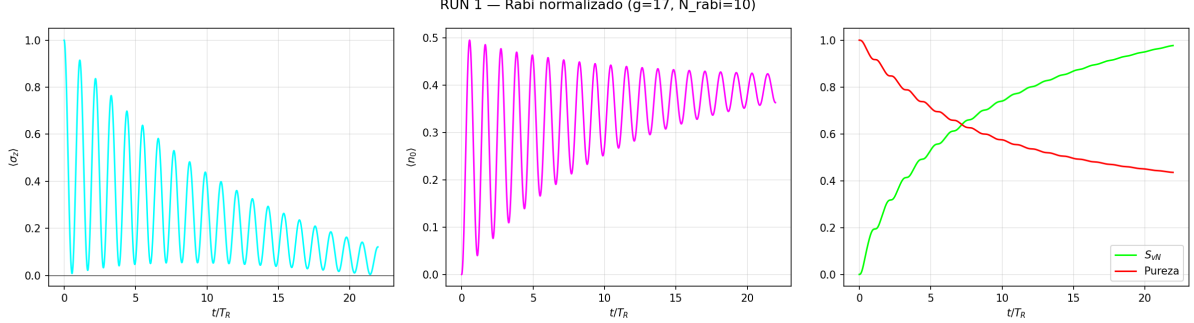


Figure 1: Rabi oscillation for $g = 17$ Hz, $N_{\text{Rabi}} = 10$ periods. Left: $\langle \sigma_z \rangle(t)$. Center: $\langle n_0 \rangle(t)$. Right: entropy and purity.

3.3 Endogenous vs exogenous regimes

We compare three cases with equal $\mathcal{R} = 10$ but different individual values of g and γ_ϕ :

- **Endogenous:** $g = 5$, $\gamma_\phi = 0.40$.
- **Mixed:** $g = 17$, $\gamma_\phi = 1.36$.
- **Exogenous:** $g = 50$, $\gamma_\phi = 4.90$.

Table 1 reports the steady-state observables obtained after $N_{\text{Rabi}} = 30$ periods.

Table 1: Steady-state observables for three regimes with $\mathcal{R} = 10$.

Regime	$\langle \sigma_z \rangle_{ss}$	$\langle n_0 \rangle_{ss}$	$S_{vN,ss}$	Purity _{ss}
Endogenous	+0.402	0.199	0.933	0.517
Mixed	-0.296	0.367	1.213	0.317
Exogenous	-0.166	0.435	1.118	0.360

The three regimes are qualitatively distinct despite equal $\mathcal{R}$. The endogenous case remains analytic-dominated ($\langle \sigma_z \rangle > 0$) with minimal substrate excitation. The exogenous case saturates the substrate ($\langle n_0 \rangle_{ss} = 0.435$). This demonstrates that $\mathcal{R}$ is *not* the unique control parameter.

3.4 Phase diagram in the (α, γ_ϕ) plane

We sweep $\alpha \in [0.05, 2.0]$ and $\gamma_\phi \in [0.05, 20]$ Hz, with $\delta = 34$ Hz fixed. Figure 2 shows the four steady-state observables. Table 2 reports the boundary $\alpha_{\text{cruce}}(\gamma_\phi)$ at which $\langle \sigma_z \rangle_{ss}$ changes sign.

The boundary is *not* a horizontal line: it is a curve. For $\gamma_\phi < 1$ Hz, the empirical scaling

$$g_{\text{cruce}}(\gamma_\phi) \cdot \gamma_\phi^{0.62} \approx 3.95 \quad (13)$$

provides an adequate fit. For $\gamma_\phi > 5$ Hz, the boundary saturates at the lower edge of the swept range, indicating that strong dissipation drives the system into the substrate-dominated regime regardless of α .

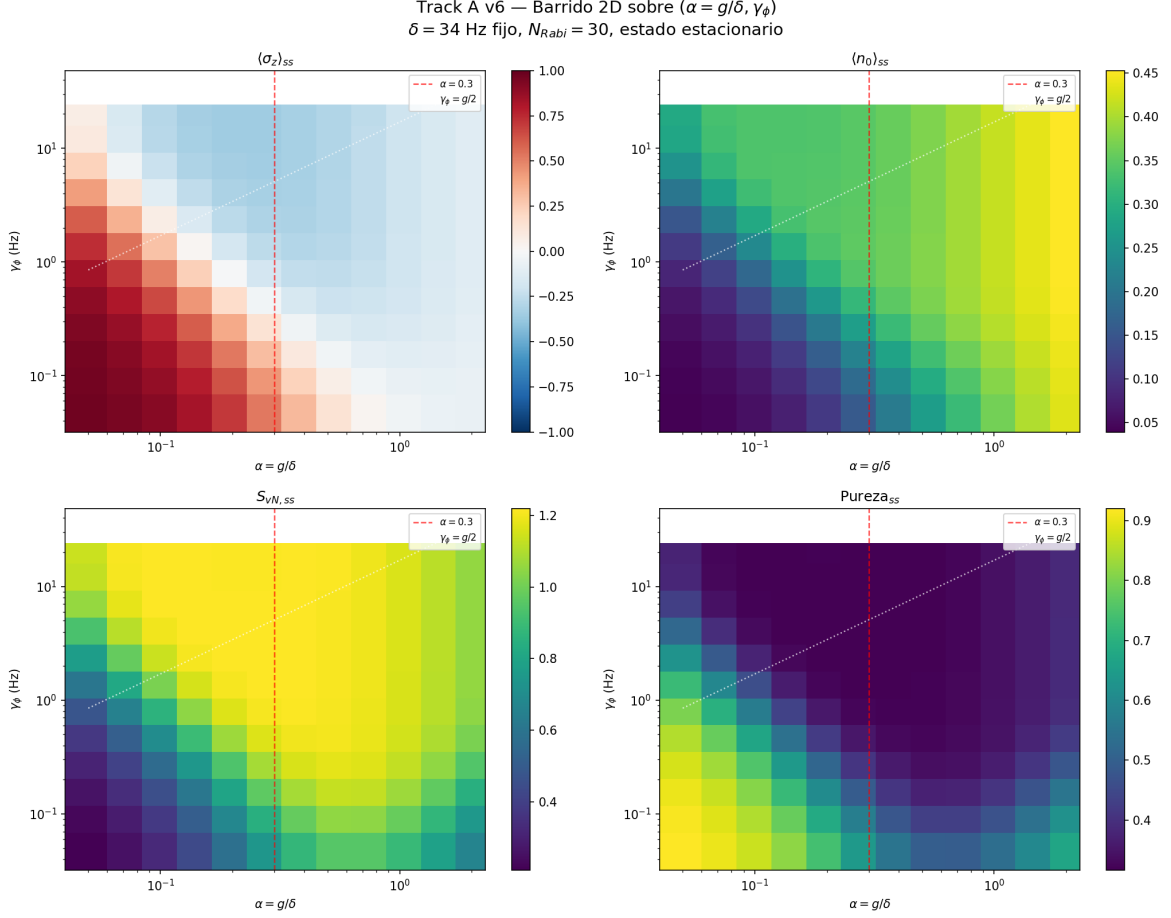


Figure 2: Phase diagram in the (α, γ_ϕ) plane. Red dashed line: $\alpha = 0.3$. White dotted line: $\gamma_\phi = g/2$.

3.5 One-dimensional cuts

Figure 3 shows cuts at $\gamma_\phi \in \{0.09, 0.44, 2.26, 11.60\}$ Hz. The transition is smooth, and its location depends on γ_ϕ . This confirms that a single-parameter threshold does not capture the dynamics.

4 Falsifiable predictions

The model produces four falsifiable predictions for EEG/fMRI experiments.

1. **Endogenous access produces low substrate excitation.** A subject trained to enter a coherent state voluntarily (meditative absorption, flow) should show: (i) stable or reduced BOLD entropy in cortical networks, (ii) increased intra-DMN phase-locking value, (iii) decreased inter-network coupling with sensory areas. In the model, this corresponds to $\langle n_0 \rangle_{ss} \approx 0.2$ and $\langle \sigma_z \rangle_{ss} > 0$.
2. **Exogenous access produces substrate saturation.** A subject under pharmacological intervention (classical psychedelics) should show: (i) increased global BOLD entropy, (ii) decreased modularity, (iii) collapse of intra-DMN connectivity. In the model, this corresponds to $\langle n_0 \rangle_{ss} \approx 0.4$ and $\langle \sigma_z \rangle_{ss} < 0$.

Table 2: Boundary α_{cruce} as a function of γ_ϕ .

γ_ϕ (Hz)	α_{cruce}	g_{cruce} (Hz)
0.050	0.731	24.9
0.086	0.523	17.8
0.149	0.374	12.7
0.256	0.267	9.1
0.442	0.191	6.5
0.762	0.137	4.7
1.313	0.137	4.7
2.264	0.098	3.3
3.903	0.070	2.4
6.729	0.050	1.7
20.00	0.050	1.7

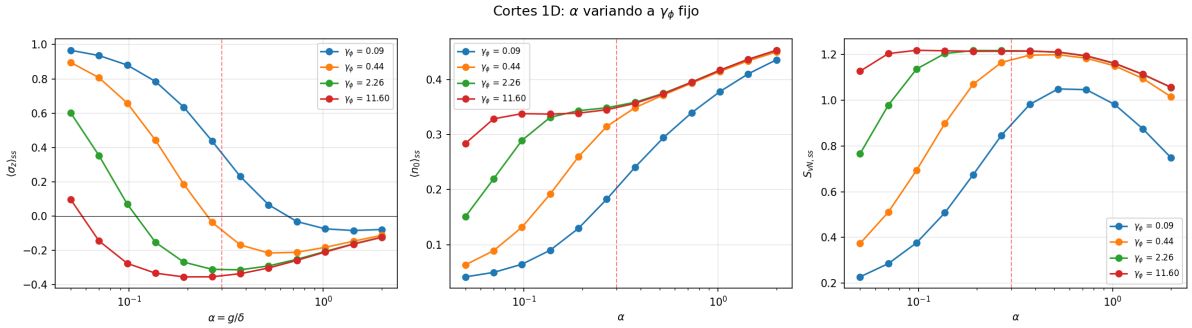


Figure 3: One-dimensional cuts at fixed γ_ϕ .

3. **The boundary is a curve, not a threshold.** The transition between analytical dominance and substrate dominance should occur at different effective coupling values depending on the subject's baseline decoherence rate. Subjects with high baseline γ_ϕ (anxious, sleep-deprived) should require lower coupling to reach the substrate regime.
4. **The steady state distinguishes endogenous from exogenous at equal $\mathcal{R}$.** If two subjects enter the substrate regime with the same ratio $\mathcal{R} = g/(\gamma_\phi + \gamma_a)$ but different absolute values, their steady states should differ in $\langle n_0 \rangle_{ss}$ by a factor of two or more.

5 Discussion

5.1 The failure of g_{crit}

Earlier proposals [8, 9] suggested $g_{\text{crit}} = \delta/2$ as a threshold for substrate access. Our simulations show that this is a geometric label of the mixing angle $\theta_0 = \pi/8$, not a dynamical transition. The system exhibits smooth crossover at a boundary that depends on both α and γ_ϕ .

5.2 What the model does and does not claim

The model is a phenomenological description of two coupled modes with dissipation. It does not claim that the brain is a quantum system in the physical sense. It claims that the *mathematical structure* of two-mode coupling with Lindblad dissipation can capture the qualitative distinction between two modes of information processing.

5.3 Limitations

The model uses a single effective DMN signal. Real neural dynamics involve many coupled populations. The mean-field approximation behind the DMN modulation is a simplification. The choice of $\omega_0 = 6$ Hz and $\omega_1 = 40$ Hz is arbitrary and should be fitted to data.

6 Conclusion

We have presented a two-mode Jaynes–Cummings model of endogenous and exogenous phase coherence. The model predicts that the boundary between analytical and substrate dominance is a curve in the (α, γ_ϕ) plane, not a single threshold. It further predicts that endogenous and exogenous substrate access differ in steady-state substrate excitation, even at equal coupling-to-dissipation ratio. Both predictions are falsifiable with current EEG/fMRI techniques.

References

- [1] M. Breakspear, *Dynamic models of large-scale brain activity*, Nat. Neurosci. **20**, 340 (2017).
- [2] Y. Kuramoto, *Self-entrainment of a population of coupled non-linear oscillators*, in *International Symposium on Mathematical Problems in Theoretical Physics* (1975).
- [3] W. J. Freeman, *Neurodynamics: An Exploration in Mesoscopic Brain Dynamics*, Springer (2000).
- [4] H. Umezawa, *Advanced Field Theory: Micro, Macro and Thermal Physics*, AIP Press (1993).
- [5] E. T. Jaynes and F. W. Cummings, *Comparison of quantum and semiclassical radiation theories with application to the beam maser*, Proc. IEEE **51**, 89 (1963).
- [6] G. Lindblad, *On the generators of quantum dynamical semigroups*, Commun. Math. Phys. **48**, 119 (1976).
- [7] C. W. Gardiner and P. Zoller, *Quantum Noise*, 3rd ed., Springer (2004).
- [8] M. A. Percudani and J. I. Díaz, *The Binary Universe Ontology: Antifrequency Validation and the Geometric Origin of the k_{UAT} Coupling Factor*, Zenodo (2025). <https://doi.org/10.5281/zenodo.18015439>
- [9] M. A. Percudani, *Anti-frequency evaluated through the UAT temporal framework*, Zenodo (2025). <https://doi.org/10.5281/zenodo.17516256>